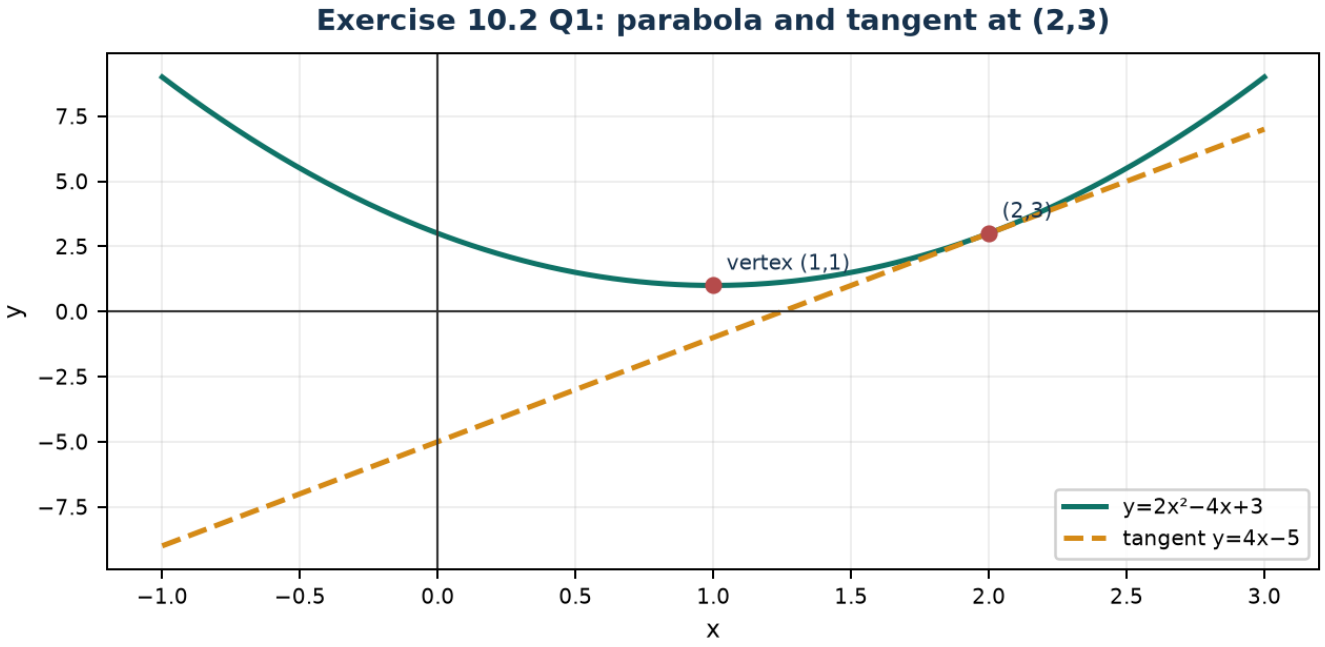
Chapter 10 • Graphs of Functions

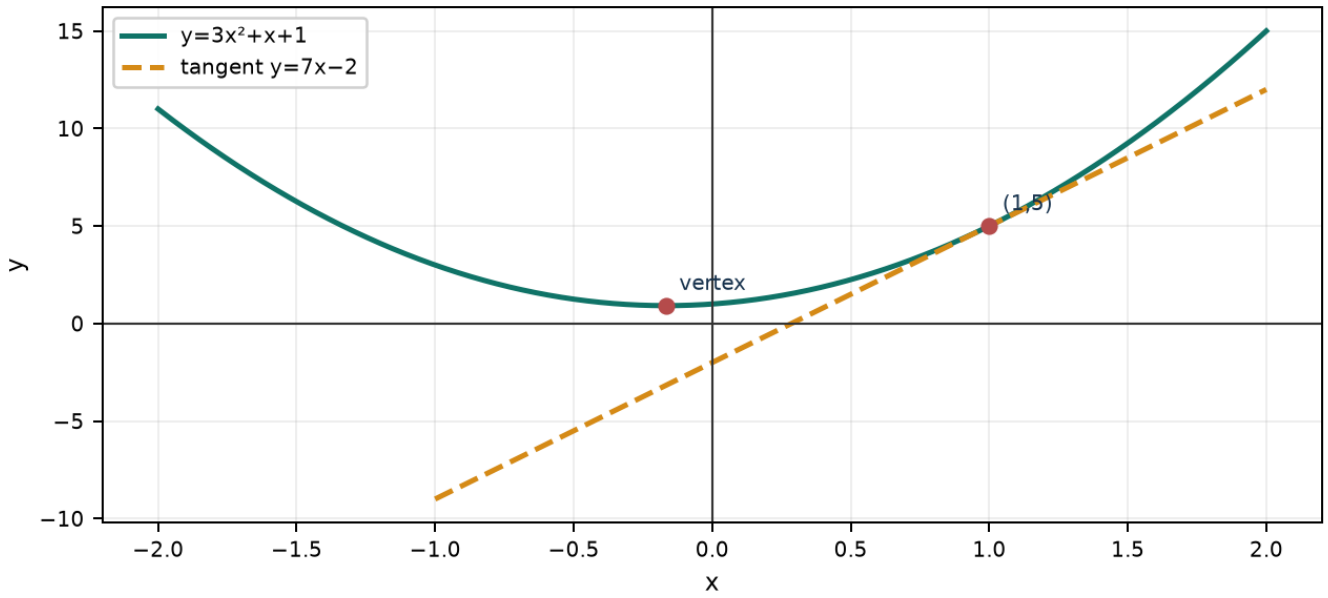
Solved Exercise 10.2

x	y
-1	9
0	3
1	1
2	3
3	9



x	y
-2	11
-1	3
-0.1667	0.9167
0	1
1	5
2	15

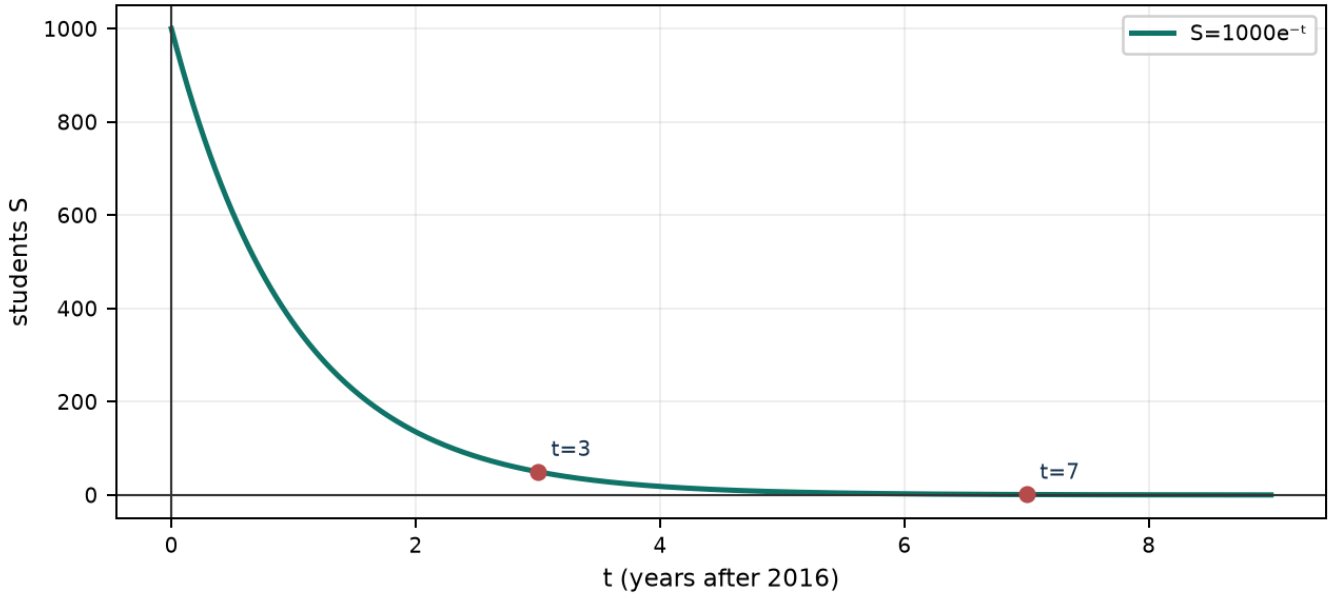
Exercise 10.2 Q2: parabola and tangent at (1,5)



t (years after 2016)	$S=1000e^{-t}$
0	1,000
1	367.8794
3	49.7871
5	6.7379
7	0.9119
9	0.1234

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Exercise 10.2 Q3(a): student-strength decay



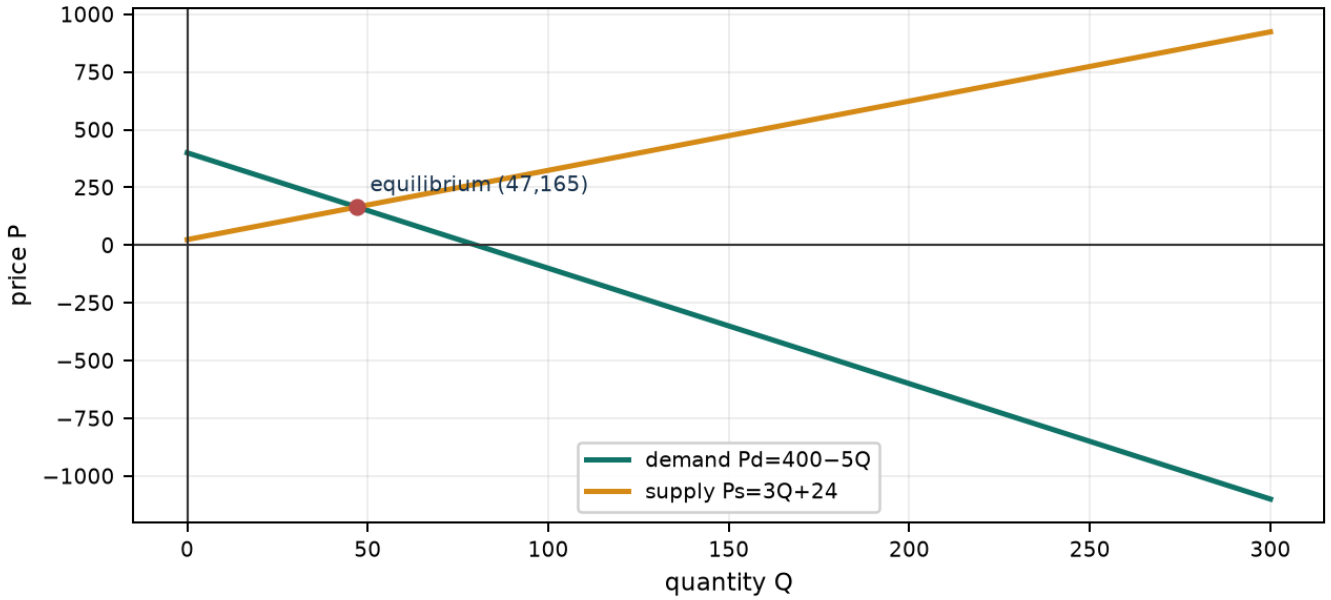
3(b). From the graph and formula in Q3(a), estimate the student strength in 2019 and in 2023.

For 2019, $t=2019-2016=3$: $S=1000e^{-3}=49.787068\ldots$
For 2023, $t=2023-2016=7$: $S=1000e^{-7}=0.911882\ldots$

Rounding to whole students gives 50 and 1 respectively.
Approximately 50 students in 2019 and 1 student in 2023.

Q	Pd	Ps
0	400	24
47	165	165
80	0	264
150	-350	474
300	-1,100	924

Exercise 10.2 Q4: supply-demand equilibrium



years of service x	salary S (Rs.)
0	45,000
1	49,500
2	54,000
3	58,500
4	63,000
5	67,500



6(a). For C(x)=1200+20x and R(x)=50x, find the break-even point.

At break-even, revenue equals cost: R(x)=C(x).
50x=1200+20x
⇒ 30x=1200
⇒ x=40.
At x = 40, R
= C
= 50(40)
= Rs. 2,000.
Break-even=(40 bags,Rs. 2,000).

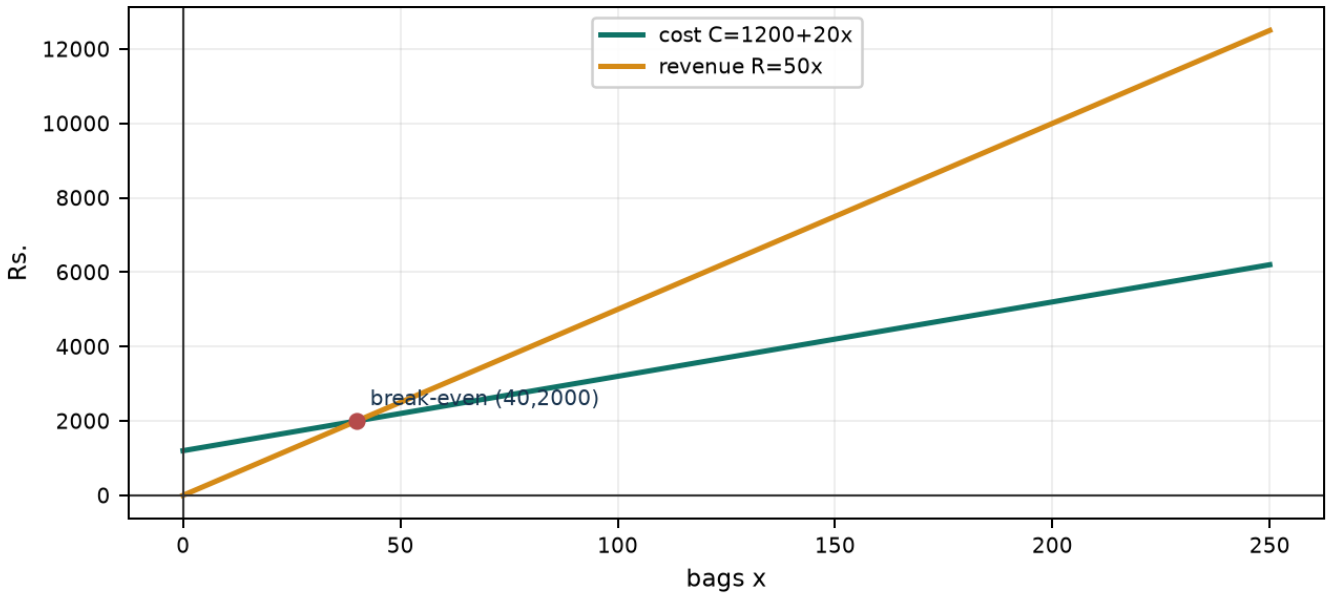
6(b). For the same school-bag model, determine the profit or loss when 250 bags are sold.

Revenue: R(250) = 50(250)
= Rs. 12,500.
Cost: C(250) = 1200+20(250)
= Rs. 6,200.
Profit = R−C
= 12,500−6,200
= Rs. 6,300.
Profit at 250 bags=Rs. 6,300.

bags x	C(x) (Rs.)	R(x) (Rs.)
0	1,200	0
20	1,600	1,000
40	2,000	2,000
60	2,400	3,000
80	2,800	4,000
250	6,200	12,500

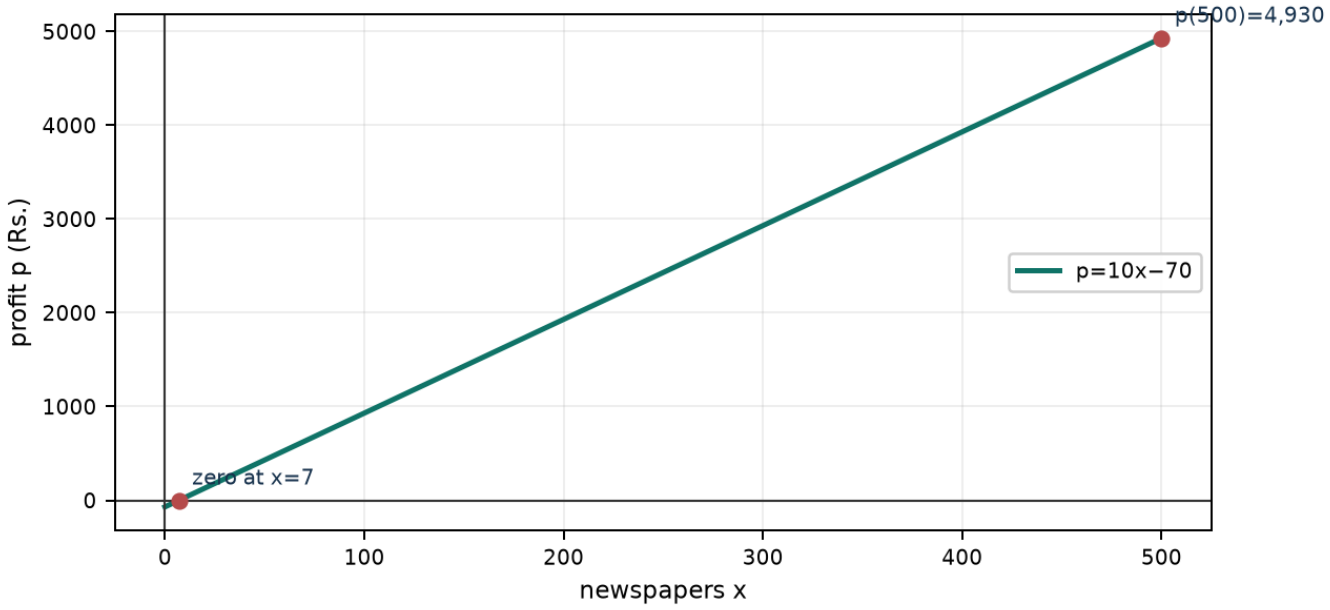
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Exercise 10.2 Q6(c): bag cost and revenue



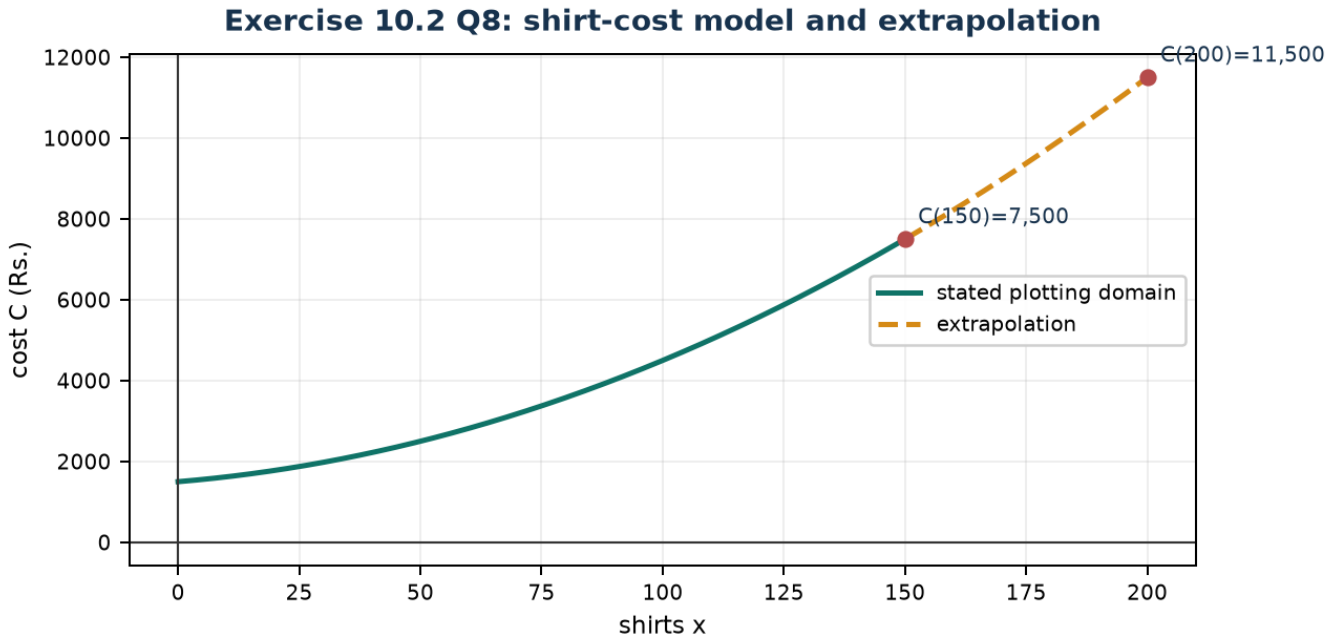
newspapers x	profit p(x) (Rs.)
0	-70
7	0
8	10
100	930
500	4,930

Exercise 10.2 Q7: newspaper profit



shirts x	cost C(x) (Rs.)
0	1,500
30	1,980

shirts x	cost C(x) (Rs.)
60	2,820
90	4,020
120	5,580
150	7,500
200	11,500



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